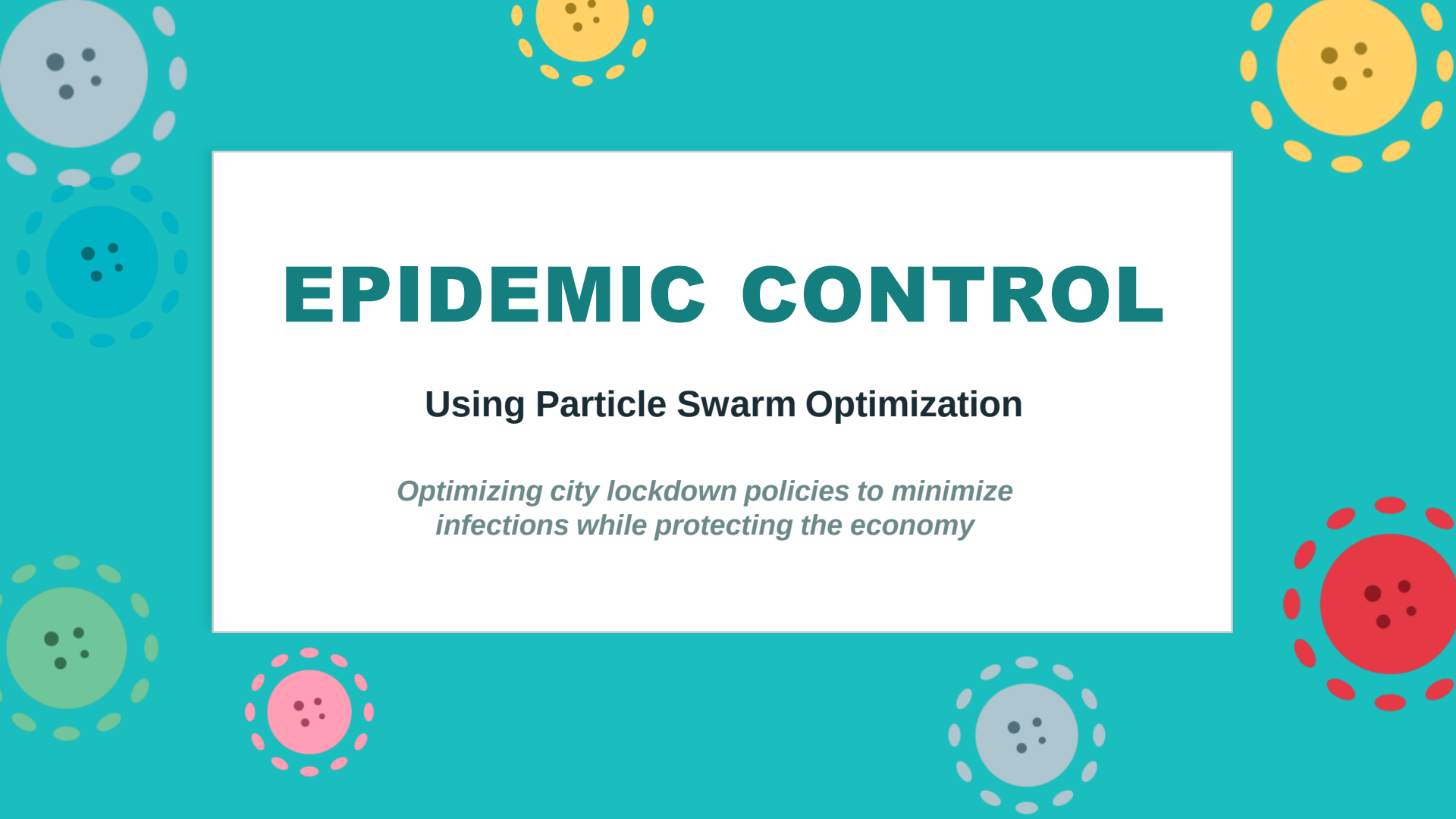




EPIDEMIC CONTROL

Using Particle Swarm Optimization

Optimizing city lockdown policies to minimize infections while protecting the economy



THE PROBLEM

SCENARIO

- A COVID-like outbreak is spreading across a city with 8 distinct zones
- Each zone has different population, risk level, and economic importance
- The government must decide: how strictly to restrict each zone?

Too strict → **economy collapses**

Too lenient → **hospitals overwhelmed**

WHY PSO?



Brute Force

Infinite combinations — impossible



Genetic Algorithm

Designed for discrete problems



PSO

Built for continuous real-valued search spaces

The optimal policy is not obvious — PSO tries to find it.





CITY ZONES

Problem Setup

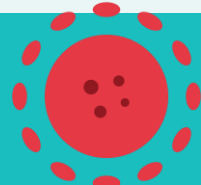
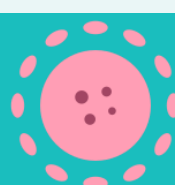
Zone	Population	Risk	Econ Val	Notes
City Center	10,000	1.5	1.5	High density, offices, transport
North District	8,000	1.0	1.0	Average residential area
East District	12,000	1.1	0.9	Large suburb, moderate risk
South District	6,000	0.9	0.8	Lower density, quieter area
West District	9,000	1.0	1.1	Mixed commercial/residential
University Area	15,000	1.7	1.2	High mixing, student dorms
Industrial Zone	7,000	0.8	2.0	Factories — very expensive!
Suburb North	5,000	0.6	0.5	Low density, low risk

Risk:

High

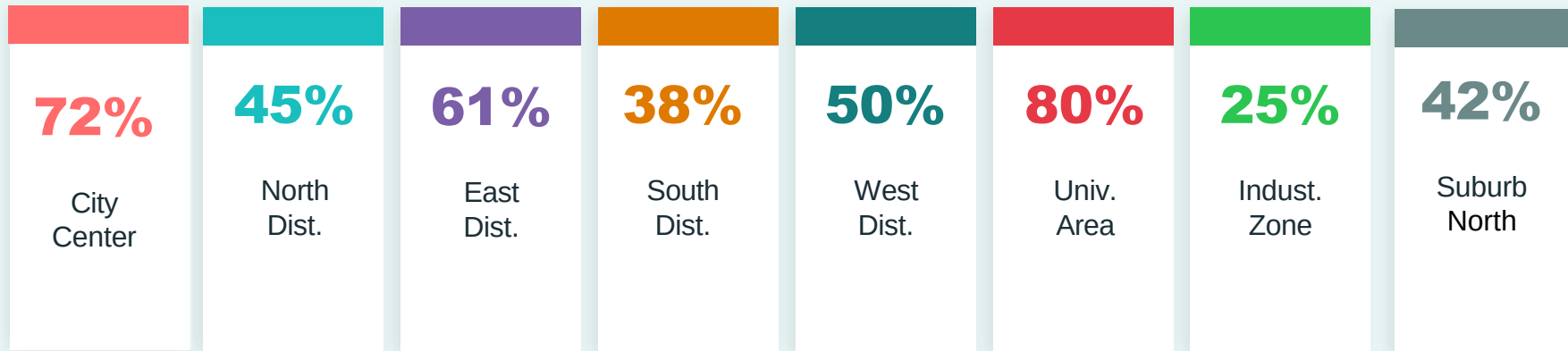
Medium

Low



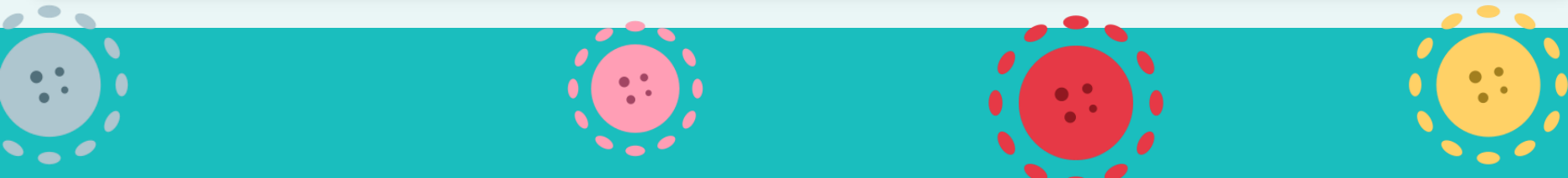
STEP 1 — Representation

Each particle = one complete city-wide restriction policy



velocity (starts at 0.0 — particles are stationary before first move):

[0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0]





STEP 2 — Fitness Function

$$\text{fitness} = (0.5 \times \text{norm_infected} + 0.5 \times \text{norm_cost})$$



Effective R

$\text{eff_R} = R_0 \times \text{risk} \times (1 - \text{restriction})$

$R_0=2.5$ is real COVID-19 value.

risk adjusts per zone density.

$(1 - \text{restriction})$ = fraction of contacts remaining.



Attack Rate

$\text{attack_rate} = 1 - (1/\text{eff_R})$ if $\text{eff_R} > 1$

Fraction of population eventually infected.

From real epidemiology (herd immunity formula).

If $\text{eff_R} \leq 1 \rightarrow$ disease dying out $\rightarrow$ rate = 0.



Residual Spread

$\text{infected} = \text{pop} \times (\text{attack_rate} + 0.02)$

0.02 = 2% always infected regardless.

Essential workers, household transmission.

Mirrors real COVID — zero infections impossible.



Economic Cost

$\text{cost} = \text{restriction} \times \text{pop} \times 100 \times \text{econ_val}$

100 = Money lost per person per 100% restriction.

$\text{econ_val} = 2$ for Industrial Zone $\rightarrow$ 2× more expensive.

PSO looks for the minimum fitness $\rightarrow$ fewer infections + less damage = better policy



STEP 3 — Velocity & Position Update

$$v_{\text{new}} = w \cdot v_{\text{old}} + c1 \cdot r1 \cdot (pbest - pos) + c2 \cdot r2 \cdot (gbest - pos)$$

→ Inertia

$$w \times v_{\text{old}} \quad (w = 0.5)$$

- Keeps particle moving in the same direction. Resistance to movement.
- Higher inertia (0.8–1) → more exploration
- Lower inertia (0.1–0.4) → more exploitation
- Ours (0.5) balances both

↖ Cognitive

$$c1 \times r1 \times (pbest - pos) \quad (c1 = 1.5)$$

- Pulls particle back toward its own personal best. Promotes individual exploration.
- Cognitive constant between 0 and 2
- Higher $c1$ = more individual exploration

★ Social

$$c2 \times r2 \times (gbest - pos) \quad (c2 = 1.5)$$

- Pulls particle toward the swarm's global best.
- Social acceleration is determined by using a constant and scaling it with a random number.
- $R2$ encourages diversity in favoring the social factor.

$$position_{\text{new}} = position_{\text{old}} + velocity_{\text{new}} \quad \text{then clip to } [0.0, 1.0]$$

STEP 4 — pbest, gbest & Convergence

pbest — Personal Best

Each particle remembers the best position it has ever personally visited.

If new fitness < pbest fitness → update pbest

gbest — Global Best

The single best position found by ANY particle in the entire swarm ever.

If any particle improves → gbest updates → ALL particles immediately pull toward it.

This is the "swarm intelligence" — collective knowledge sharing.

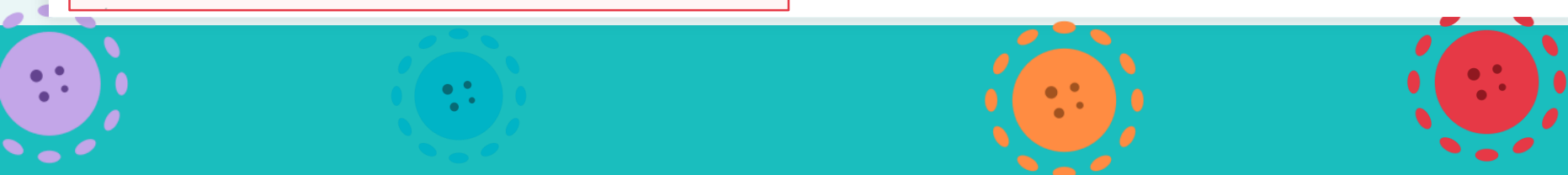
Convergence — When Does PSO Stop?

```
if particle.fit < gbest_fit + 1e-7:    meaningful improvement → reset stall counter (+ 1e-7 to ignore small fractional changes)
```

```
if stall_count >= 40:    no improvement for 40 consecutive iterations → STOP
```

Iteration 0 | fitness: 0.384923 ← starts bad (particles near extremes)

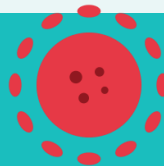
Iteration 20 | fitness: 0.331232 ← improving, swarm converging



RESULTS — Optimal Policy Found

💡 Key Insight: University Area (risk 1.5, average loss) → highest restriction.
Industrial Zone (risk 0.8, very expensive) → no restriction.

ZONE	ORIGINAL POP	RESTR.	INFECTED	ECON COST
City Center	10,000	73.3%	200	1100001
North District	8,000	60.0%	160	480000
East District	12,000	63.6%	240	687274
South District	6,000	55.6%	120	266667
West District	9,000	60.0%	180	594000
University Area	15,000	76.5%	300	1376471
Industrial Zone	7,000	0.0%	3640	0
Suburb North	5,000	33.3%	100	83334
TOTAL	72,000	-	4,940	4,587,745



PSO SUMMARY

Epidemic Control — Key Takeaways

01

Representation: Each particle = one 8-zone restriction policy (real values 0–1)

02

Fitness: Minimize infections + cost using eff_R, attack rate, residual spread

03

Velocity: $w=0.5$, $c1=1.5$, $c2=1.5$

04

pbest & gbest: Personal memory + swarm knowledge guide all 50 particles every iteration

05

Convergence: Stops after 40 iterations of no meaningful improvement (stall limit)

PSO efficiently finds the optimal policy without checking all combinations — this is its core advantage over brute force search